

Two Apéry-transform supercongruences in OEIS A267220

A coefficient proof, a parameter refinement,
and the dyadic correction

Research note prepared for Vladimir Reshetnikov

September 20, 2026

Abstract

Two formulas contributed by Peter Bala to OEIS A267220 on October 17, 2024 assert second-order supercongruences for diagonal powers of the exponential transform of the classical Apéry numbers and of its associated reversion transform. Both formulas are true for every integer value of the parameter. We prove them by establishing the second-order Gauss congruences for the Apéry numbers directly from their binomial sum, then giving a coefficient-level proof of preservation under normalized framing. For odd primes the result gains an additional factor of the prime dividing the parameter, and the restriction to primes at least five can be removed. The second family satisfies the original unsigned congruence also at the prime two; the first family requires an explicit sign correction at the first dyadic descent. We prove the full signed statement, treat zero and negative parameters, and give exact counterexamples to two tempting stronger assertions. The general framing-integrality mechanism is established prior work of Schwarz, Vologodsky, and Walcher, not a novelty claim of this note. The purpose here is to resolve the specific OEIS assertions with a complete, directly checkable argument. A standard-library Python program and its exact verification records accompany the article.

1 The two conjectures and the result

Let

$$a_n = \sum_{k=0}^n \binom{n}{k}^2 \binom{n+k}{k}^2 \quad (n \geq 0) \quad (1.1)$$

be the classical Apéry numbers, OEIS A005259 [2]. Their initial values are

$$1, 5, 73, 1445, 33001, 819005, 21460825, \dots$$

The two series relevant to the problem are

$$A(x) = \exp\left(\sum_{n \geq 1} \frac{a_n}{n} x^n\right), \quad F(x) = \frac{x}{\text{Rev}(xA(x))}. \quad (1.2)$$

Here Rev denotes compositional inversion at zero. Since $xA(x) = x + O(x^2)$, this inverse exists uniquely over $\mathbb{Q}[[x]]$. The quotient defining F is a well-defined formal power series with constant coefficient one. Integrality of both series will be proved below; it is not assumed in the argument.

The coefficients of A form OEIS A267220. The entry includes two conjectures, both explicitly attributed to Peter Bala and dated October 17, 2024 [1]. In our notation they concern

$$u_m(n) = [x^n]A(x)^{mn}, \quad v_m(n) = [x^n]F(x)^{mn} \quad (m \in \mathbb{Z}, n \geq 0), \quad (1.3)$$

and assert

$$\begin{aligned} u_m(np^r) &\equiv u_m(np^{r-1}) \pmod{p^{2r}}, \\ v_m(np^r) &\equiv v_m(np^{r-1}) \pmod{p^{2r}} \end{aligned} \quad (1.4)$$

for every prime $p \geq 5$ and all positive integers n, r . These assertions were still labeled conjectural when the entry was checked on September 20, 2026.

The notation a_n in this article always denotes the Apéry numbers, *not* the coefficients of A267220. In particular, these are two related but different sequences. Also,

$$u_m(0) = v_m(0) = 1$$

for every m , whereas $u_0(n) = v_0(n) = 0$ for $n \geq 1$. This distinction matters when normalizing by the parameter.

Theorem 1.1 (Resolution and refinements). *Both congruences (1.4) hold for every integer m . More precisely, let $N \geq 1$, let $p \mid N$ be prime, and put $R = \nu_p(N) \geq 1$.*

(a) *If p is odd and $m \neq 0$, then*

$$\begin{aligned} \nu_p(u_m(N) - u_m(N/p)) &\geq 2R + \nu_p(m), \\ \nu_p(v_m(N) - v_m(N/p)) &\geq 2R + \nu_p(m). \end{aligned} \quad (1.5)$$

(b) *The unsigned congruence for the second family holds at every prime, including 2:*

$$v_m(N) \equiv v_m(N/p) \pmod{p^{2R}}. \quad (1.6)$$

(c) *For every prime p and $m \neq 0$, the signed refinements are*

$$\begin{aligned} (-1)^{mN} u_m(N) &\equiv (-1)^{mN/p} u_m(N/p) \pmod{p^{2R+\nu_p(m)}}, \\ (-1)^{(m-1)N} v_m(N) &\equiv (-1)^{(m-1)N/p} v_m(N/p) \pmod{p^{2R+\nu_p(m)}}. \end{aligned} \quad (1.7)$$

For $m = 0$, all the positive-index terms in these statements vanish.

We use $\nu_p(0) = +\infty$, and $\nu_p(m) = \nu_p(|m|)$ for nonzero negative integers. Substituting $N = np^r$ gives $R = r + \nu_p(n)$, so part (a) proves a modulus at least as strong as the one in OEIS even when p divides n . The theorem does not tacitly assume $p \nmid n$ or $p \nmid m$.

What is established here, and what is prior work?

The nonlinear preservation principle behind the proof is the *integrality of framing* for 2-functions. It is proved in general by Schwarz, Vologodsky, and Walcher [3, 4]; their 2017 paper includes a local coefficient proof. Our proof uses that known mechanism in an explicit normalization adapted to the two OEIS families. No claim is made that the general preservation theorem, or its connection to dilogarithms, is new. Nor does a conjectural label in OEIS establish that no earlier specialist has deduced this particular application.

The contribution of this note is a self-contained resolution of the stated formulas, with all parameter values and the prime two handled explicitly, together with reproducible computations. The proof does not appeal to a computer search as a substitute for an all-index argument. Its dependency chain is

$$\begin{aligned} \text{Apéry binomial sum} &\implies \text{second-order Gauss congruences} \implies \text{integral } A, \\ \text{normalized coefficient transform} &\implies \text{framing congruences} \implies \text{both OEIS assertions.} \end{aligned}$$

2 Formal arithmetic and the dilogarithm criterion

Everything below takes place in formal power series rings. No analytic convergence or choice of a complex logarithm is involved. Set

$$\delta = x \frac{d}{dx}, \quad \text{CT} \left(\sum_{j \geq j_0} q_j x^j \right) = q_0.$$

The constant-term functional is applied only to Laurent series bounded below. It satisfies

$$\text{CT}(\delta G) = 0, \quad \text{CT}(G \delta J) = -\text{CT}(J \delta G). \quad (2.1)$$

Each relevant constant term depends on only finitely many coefficients.

For a prime p , write

$$\mathbb{Z}_{(p)} = \{a/b \in \mathbb{Q} : p \nmid b\}.$$

A congruence between elements of $\mathbb{Z}_{(p)}$ has its usual valuation meaning; a congruence between series means the congruence of every coefficient. Working in this localization is sufficient: completeness is unnecessary.

Definition 2.1. An integer sequence $(\alpha_n)_{n \geq 1}$ satisfies the *second-order Gauss congruences* if

$$\alpha_N \equiv \alpha_{N/p} \pmod{p^{2\nu_p(N)}} \quad (2.2)$$

for every prime p and every positive N divisible by p .

Lemma 2.2 (Möbius and dilogarithm criterion). *For an integer sequence $(\alpha_n)_{n \geq 1}$, the following are equivalent:*

(i) *The sequence satisfies (2.2).*

(ii) *The numbers*

$$c_n = \frac{1}{n^2} \sum_{d|n} \mu(n/d) \alpha_d \quad (2.3)$$

are integers for every $n \geq 1$.

(iii) *There is an expansion with integer coefficients*

$$\sum_{n \geq 1} \frac{\alpha_n}{n^2} x^n = \sum_{d \geq 1} c_d \text{Li}_2(x^d), \quad \text{Li}_2(z) = \sum_{j \geq 1} \frac{z^j}{j^2}. \quad (2.4)$$

The c_d in (iii) are uniquely given by (2.3).

Proof. Fix $n = p^r h$ with $p \nmid h$. In the numerator of (2.3), the only nonzero terms have divisor p -adic exponent r or $r - 1$. Pairing them gives

$$\sum_{d|n} \mu(n/d) \alpha_d = \sum_{e|h} \mu(h/e) (\alpha_{p^r e} - \alpha_{p^{r-1} e}).$$

Under (i), this numerator is divisible by p^{2r} . Doing this for every prime divisor of n proves divisibility by n^2 , hence (ii).

Möbius inversion gives

$$\alpha_n = \sum_{d|n} d^2 c_d. \quad (2.5)$$

If every c_d is an integer, subtracting the formula for $\alpha_{N/p}$ from that for α_N leaves only divisors d with $\nu_p(d) = \nu_p(N)$. Every surviving term contains $p^{2\nu_p(N)}$, proving (i). Finally, coefficient extraction from the right side of (2.4) gives $n^{-2} \sum_{d|n} d^2 c_d$, proving the equivalence with (iii). $\square$

One immediate benefit of this stronger integrality criterion is an Euler product. Define

$$W(x) = \sum_{n \geq 1} \frac{\alpha_n}{n^2} x^n, \quad H(x) = \delta W(x) = \sum_{n \geq 1} \frac{\alpha_n}{n} x^n.$$

Taking δ of (2.4) and then exponentiating gives

$$\exp H(x) = \prod_{d \geq 1} (1 - x^d)^{-d c_d} \in 1 + x \mathbb{Z}[[x]]. \quad (2.6)$$

This identity is formal: to compute the coefficient of x^n , only factors with $d \leq n$ are needed. The exponents may be negative or positive; any integer power of a unit in $1 + x \mathbb{Z}[[x]]$ is integral.

A second formulation will be central to the coefficient proof.

Lemma 2.3 (Integral Frobenius defect). *If (α_n) satisfies the second-order Gauss congruences, then for every prime p the series*

$$E_p(x) = W(x) - \frac{1}{p^2} W(x^p) \quad (2.7)$$

belongs to $x \mathbb{Z}_{(p)}[[x]]$.

Proof. If $p \nmid n$, the coefficient is $\alpha_n/n^2 \in \mathbb{Z}_{(p)}$. If $p \mid n$, it is $(\alpha_n - \alpha_{n/p})/n^2$, again in $\mathbb{Z}_{(p)}$ by the hypothesis. $\square$

3 The required Apéry congruence from the binomial sum

We now return to the sequence a_n in (1.1). Only a second-order congruence is needed. It admits a short proof that works uniformly at all primes, including 2 and 3.

Proposition 3.1. *For every prime p and positive integer N divisible by p ,*

$$a_N \equiv a_{N/p} \pmod{p^{2\nu_p(N)}}. \quad (3.1)$$

Proof. Put $R = \nu_p(N)$ and

$$P(N, k) = \binom{N}{k} \binom{N+k}{k}.$$

When $p \nmid k$, the identity

$$\binom{N}{k} = \frac{N}{k} \binom{N-1}{k-1}$$

shows that $\nu_p(P(N, k)) \geq R$. Such a summand $P(N, k)^2$ therefore vanishes modulo p^{2R} .

It remains to compare the summands with $k = p\ell$. For $k \geq 1$, an exact product identity is

$$P(N, k) = (-1)^{k-1} \frac{N}{k} \left(1 + \frac{N}{k}\right) \prod_{j=1}^{k-1} \left(1 - \frac{N^2}{j^2}\right). \quad (3.2)$$

To see this, write the binomial product as

$$\frac{N(N+k)}{k^2} \prod_{j=1}^{k-1} \frac{(N-j)(N+j)}{j^2}$$

and extract the signs. Separating the factors with $p \mid j$ in (3.2) yields

$$P(N, p\ell) = (-1)^{(p-1)\ell} P(N/p, \ell) \prod_{\substack{1 \leq j < p\ell \\ p \nmid j}} \left(1 - \frac{N^2}{j^2}\right). \quad (3.3)$$

Each factor in the remaining product belongs to $1 + p^{2R}\mathbb{Z}_{(p)}$. Squaring removes the sign even when $p = 2$, so

$$P(N, p\ell)^2 \equiv P(N/p, \ell)^2 \pmod{p^{2R}}.$$

The $k = 0$ summand is one on both sides. Summing over $0 \leq k \leq N$ now gives exactly (3.1). $\square$

Consequently, Lemma 2.2 applies to a_n , and the series A in (1.2) is integral by (2.6). Its primitive dilogarithm coefficients begin

$$c_1 = 5, \quad c_2 = 17, \quad c_3 = 160, \quad c_4 = 2058, \quad c_5 = 32760. \quad (3.4)$$

For example, $c_4 = (a_4 - a_2)/16 = 2058$. Thus

$$A(x) = (1-x)^{-5}(1-x^2)^{-34}(1-x^3)^{-480}(1-x^4)^{-8232} \dots$$

The arithmetic input to the rest of the proof is now complete. We have not used a stronger Apéry supercongruence as a black box.

4 An integral normalization of the two transforms

From now on set

$$W = \sum_{n \geq 1} \frac{a_n}{n^2} x^n, \quad H = \delta W = \log A, \quad D = \delta H = \sum_{n \geq 1} a_n x^n. \quad (4.1)$$

The series D has constant coefficient zero; in particular, it is the Apéry generating series with its initial term $a_0 = 1$ removed.

For every $m \in \mathbb{Z}$, define $X_m(y)$ by

$$X_m(y) = yA(X_m(y))^m, \quad \text{equivalently} \quad y = xA(x)^{-m}. \quad (4.2)$$

The inverse exists uniquely in $y + y^2\mathbb{Z}[[y]]$: a series $x + O(x^2)$ with integer coefficients has an integer compositional inverse, as is seen by solving successively for its coefficients. This includes negative m , since A and every integer power of A are integral units.

Define the normalized transform by

$$\mathcal{B}_m(y) = \frac{D(X_m(y))}{1 - mD(X_m(y))} = \sum_{n \geq 1} b_m(n)y^n. \quad (4.3)$$

The denominator is a unit of constant coefficient one. Therefore

$$b_m(n) \in \mathbb{Z} \quad (m \in \mathbb{Z}, n \geq 1), \quad b_0(n) = a_n. \quad (4.4)$$

This integral definition is preferable to starting with a quotient by m . In particular, it remains valid when a prime under consideration divides m .

Proposition 4.1 (Coefficient normalization). *For $m \neq 0$ and $n \geq 1$,*

$$b_m(n) = \frac{1}{m}[x^n]A(x)^{mn}. \quad (4.5)$$

Consequently, $u_m(n) = mb_m(n)$ for every integer m and every $n \geq 1$.

Proof. Logarithmic differentiation of $y = xA(x)^{-m}$ gives

$$\frac{d \log y}{d \log x} = 1 - mD(x),$$

so the chain rule implies

$$\delta_y H(X_m(y)) = \mathcal{B}_m(y). \quad (4.6)$$

The formal Lagrange inversion formula (proved in Appendix A) gives

$$[y^n]H(X_m(y)) = \frac{1}{n}[x^{n-1}]H'(x)A(x)^{mn}.$$

For $m \neq 0$, use $(A^{mn})' = mnH'A^{mn}$. The last display becomes

$$[y^n]H(X_m(y)) = \frac{1}{mn}[x^n]A(x)^{mn}.$$

Multiplication by n , as in (4.6), proves the assertion. At $m = 0$, the identity $u_0(n) = 0 = mb_0(n)$ holds separately. $\square$

The relation between the two conjectures is particularly simple after this normalization.

Proposition 4.2 (The reversion family is a shifted transform). *The series F is integral and satisfies*

$$F(y) = A(X_{-1}(y)). \quad (4.7)$$

For every integer m and every $n \geq 1$,

$$v_m(n) = mb_{m-1}(n). \quad (4.8)$$

In particular, $v_1(n) = a_n$.

Proof. Equation (4.2) with $m = -1$ says $X_{-1}(y)A(X_{-1}(y)) = y$, hence $X_{-1} = \text{Rev}(xA(x))$. Dividing by X_{-1} proves (4.7), including its integrality.

Apply Lagrange inversion to $A(X_{-1}(y))^{mn}$. For $n \geq 1$ it gives

$$[y^n]A(X_{-1}(y))^{mn} = m[x^{n-1}]A'(x)A(x)^{(m-1)n-1}. \quad (4.9)$$

If $m \neq 1$, differentiating $A^{(m-1)n}$ turns this into

$$\frac{m}{m-1}[x^n]A(x)^{(m-1)n} = mb_{m-1}(n).$$

For $m = 1$, the right side of (4.9) is $[x^{n-1}]A'/A = [x^{n-1}]H' = a_n = b_0(n)$. Thus the apparent singularity at $m = 1$ is removable, and the asserted formula holds for all m . $\square$

Remark 4.3 (Why division modulo a prime is not enough). The formula $v_m(n) = \frac{m}{m-1}u_{m-1}(n)$ for $m \neq 1$ is an identity over $\mathbb{Q}$, but it cannot safely be used to transfer a congruence by dividing by $m-1$ when $p \mid m-1$. Similarly, dividing a congruence for u_m by m may lose powers of p . The proof below establishes the congruence for the integer b_m itself. This is why no exceptional parameter classes need to be excluded.

5 The normalized framing congruence at odd primes

The next theorem is the coefficient form of the established framing-integrality principle. Its proof is included rather than invoked from the general theory [3, 4].

Theorem 5.1 (Normalized framing congruences). *Let b_m be as in (4.3). For every integer m , define*

$$\beta_m(n) = (-1)^{mn}b_m(n) \quad (n \geq 1). \quad (5.1)$$

Then β_m satisfies the second-order Gauss congruences at every prime. Equivalently, for $p \mid N$ and $R = \nu_p(N)$,

$$(-1)^{mN}b_m(N) \equiv (-1)^{mN/p}b_m(N/p) \pmod{p^{2R}}. \quad (5.2)$$

For odd p this is the ordinary unsigned congruence for b_m . At $p = 2$ the unsigned congruence also holds whenever m is even or $R \geq 2$.

We prove the odd-prime assertion in this section and the remaining cases in Section 6. These arguments use only the integer coefficients a_n and their second-order Gauss congruences. Thus the same proof establishes the theorem for any input sequence satisfying Definition 2.1.

5.1 An exact constant-term expression

Fix $p \mid N$, put $R = \nu_p(N)$, and first take $m \neq 0$. For brevity write $E = E_p$. Lemma 2.3 gives

$$E = W(x) - p^{-2}W(x^p) \in x\mathbb{Z}_{(p)}[[x]].$$

Differentiating and using $H = \delta W$ yields

$$\delta E = H(x) - p^{-1}H(x^p), \quad A(x^p) = A(x)^p \exp(-p\delta E). \quad (5.3)$$

Introduce the integral Laurent series

$$K(x) = x^{-N}A(x)^{mN} \in x^{-N}\mathbb{Z}[[x]]. \quad (5.4)$$

Since substitution $x \mapsto x^p$ turns the coefficient of $x^{N/p}$ into the coefficient of x^N , Proposition 4.1 and (5.3) give the exact identity

$$b_m(N) - b_m(N/p) = \frac{1}{m} \text{CT}(K(1 - \exp(-mN\delta E))). \quad (5.5)$$

Expanding the exponential gives

$$b_m(N) - b_m(N/p) = \sum_{k \geq 1} \frac{(-1)^{k+1} m^{k-1} N^k}{k!} \text{CT}(K(\delta E)^k). \quad (5.6)$$

Only $k \leq N$ can contribute, because δE has no constant term. Thus no question about convergence of the infinite displayed sum arises. The factor m has disappeared from the denominator of each summand: this is the crucial normalization when $p \mid m$.

5.2 The linear term gains a second power of the index

The $k = 1$ term seems initially to contain only N , not N^2 . Constant-term integration by parts supplies the missing factor. Namely,

$$\delta K = N(mD - 1)K, \quad (5.7)$$

and therefore

$$\begin{aligned} N \text{CT}(K\delta E) &= -N \text{CT}(E\delta K) \\ &= N^2 \text{CT}(K(1 - mD)E) \in p^{2R}\mathbb{Z}_{(p)}. \end{aligned} \quad (5.8)$$

All coefficients inside the final constant term are p -integral. This identity, rather than a coarse exponential estimate alone, is what produces a second-order congruence.

5.3 The nonlinear terms at odd primes

For an odd prime and $k \geq 2$,

$$\nu_p(k!) \leq k - 2. \quad (5.9)$$

For $k = 2$ this is immediate. For $k \geq 3$, Legendre's formula gives $\nu_p(k!) \leq (k - 1)/(p - 1) \leq k - 2$. Since $\nu_p(m) \geq 0$ and $R \geq 1$,

$$\begin{aligned} \nu_p\left(\frac{m^{k-1}N^k}{k!}\right) &\geq kR - (k - 2) \\ &= 2R + (k - 2)(R - 1) \geq 2R. \end{aligned}$$

Also $\text{CT}(K(\delta E)^k) \in \mathbb{Z}_{(p)}$. Every nonlinear term in (5.6) is therefore divisible by p^{2R} . Together with (5.8), this proves

$$b_m(N) \equiv b_m(N/p) \pmod{p^{2R}} \quad (p \text{ odd, } m \neq 0). \quad (5.10)$$

For $m = 0$, this is precisely Proposition 3.1. Finally, N and N/p have the same parity when p is odd, so the two signs in (5.2) agree. This proves the odd-prime part of Theorem 5.1.

6 The prime two and the exact sign correction

The odd-prime factorial bound fails at $p = 2$. Replacing it indiscriminately by the dyadic bound would lose information and would not prove the asserted modulus. There are three distinct cases. Throughout this section $E = E_2$, $K = x^{-N}A^{mN}$, and $R = \nu_2(N) \geq 1$. The case $m = 0$ is already covered by the input Apéry congruence.

6.1 Even parameters

Suppose $m \neq 0$ is even. For $k \geq 2$, the bound $\nu_2(k!) \leq k - 1$ gives

$$\nu_2\left(\frac{m^{k-1}N^k}{k!}\right) \geq (k-1) + kR - (k-1) = kR \geq 2R.$$

The linear term is handled by (5.8). Hence

$$b_m(N) \equiv b_m(N/2) \pmod{2^{2R}} \quad (m \text{ even}). \quad (6.1)$$

The signs in (5.2) are both positive in this case.

6.2 Odd parameters beyond the first dyadic descent

Suppose m is odd and $R \geq 2$. For $k \geq 3$,

$$\nu_2\left(\frac{m^{k-1}N^k}{k!}\right) \geq kR - (k-1) \geq 2R.$$

The last inequality follows from $(k-2)R \geq k-1$, valid for $R \geq 2$ and $k \geq 3$. The only problematic nonlinear term is $k=2$, whose scalar coefficient $mN^2/2$ has valuation $2R-1$.

Write $E = \sum_{j \geq 1} e_j x^j$, with $e_j \in \mathbb{Z}_{(2)}$. Reduction modulo two gives

$$(\delta E)^2 \equiv \sum_{\substack{j \geq 1 \\ j \text{ odd}}} e_j^2 x^{2j} \pmod{2}. \quad (6.2)$$

For odd j , applying CT δ to Kx^{2j} and using (5.7) yields

$$2j \text{ CT}(Kx^{2j}) = N \text{ CT}(K(1 - mD)x^{2j}). \quad (6.3)$$

Since j is odd, the right side shows $\text{CT}(Kx^{2j}) \in 2^{R-1}\mathbb{Z}_{(2)} \subseteq 2\mathbb{Z}_{(2)}$. Equation (6.2) consequently implies

$$\text{CT}(K(\delta E)^2) \in 2\mathbb{Z}_{(2)}.$$

This supplies the missing factor two in the quadratic term. The linear term again follows from (5.8), proving

$$b_m(N) \equiv b_m(N/2) \pmod{2^{2R}} \quad (m \text{ odd}, R \geq 2). \quad (6.4)$$

Here N and $N/2$ are both even, so this is also the signed congruence.

6.3 Odd parameters at the first dyadic descent

It remains to take m odd and $N = 2h$ with h odd. The correct relation is now a *sum*:

$$b_m(2h) + b_m(h) \equiv 0 \pmod{4}. \quad (6.5)$$

Using the same derivation as in (5.5), but adding instead of subtracting, we obtain

$$b_m(N) + b_m(h) = \frac{1}{m} \text{CT}(K(1 + \exp(-mN\delta E))). \quad (6.6)$$

Let $s_2(k)$ denote the number of ones in the binary expansion of k . Legendre's formula at two is

$$\nu_2(k!) = k - s_2(k).$$

As mN has valuation one, the coefficient $(-mN)^k/k!$ has valuation $s_2(k)$. Modulo four, only positive exponents k that are powers of two survive. Each surviving scalar is congruent to two modulo four, since it has valuation exactly one. Thus, coefficientwise in $\mathbb{Z}_{(2)}[[x]]$,

$$1 + \exp(-mN\delta E) \equiv 2 \left(1 + \sum_{s \geq 0} (\delta E)^{2^s} \right) \pmod{4}. \quad (6.7)$$

The sum on the right is locally finite in degree.

We next identify its reduction modulo two. In δE , terms of even degree vanish modulo two. For odd j , $e_j = a_j/j^2$, and hence

$$\delta E \equiv \sum_{j \text{ odd}} a_j x^j \pmod{2}.$$

Frobenius in characteristic two gives

$$\sum_{s \geq 0} (\delta E)^{2^s} \equiv \sum_{s \geq 0} \sum_{j \text{ odd}} a_j x^{j2^s} \pmod{2}.$$

The input congruences imply $a_{j2^s} \equiv a_j \pmod{2}$. Every positive integer is uniquely $j2^s$ with j odd. Therefore

$$\sum_{s \geq 0} (\delta E)^{2^s} \equiv D(x) \pmod{2}. \quad (6.8)$$

Combining (6.6)–(6.8) gives

$$b_m(N) + b_m(h) \equiv \frac{2}{m} \text{CT}(K(1 + D)) \pmod{4}.$$

Because m is odd, $1 + D \equiv 1 - mD \pmod{2}$. But (5.7) and $\text{CT} \delta K = 0$ give the exact identity

$$\text{CT}(K(1 - mD)) = 0. \quad (6.9)$$

It follows that $\text{CT}(K(1 + D))$ is even, proving (6.5). Since $(-1)^{mN} = 1$ and $(-1)^{mh} = -1$, this is exactly the missing case of (5.2). Theorem 5.1 is proved.

7 Deduction of both OEIS conjectures

We now prove Theorem 1.1 and make clear where the two families behave differently at two.

7.1 Odd primes and the parameter factor

For p odd, Theorem 5.1 and the identities

$$u_m(n) = mb_m(n), \quad v_m(n) = mb_{m-1}(n)$$

immediately imply

$$\begin{aligned} u_m(N) - u_m(N/p) &\in mp^{2R}\mathbb{Z}_{(p)}, \\ v_m(N) - v_m(N/p) &\in mp^{2R}\mathbb{Z}_{(p)}. \end{aligned}$$

This is (1.5). In particular, the originally stated range $p \geq 5$ may be replaced by all odd primes.

Multiplying the all-prime signed congruence for b_m by m proves the first part of (1.7). Applying the same congruence to b_{m-1} and multiplying by m proves its second part. Notice that the sign parameter for v_m is $m - 1$, but the extra valuation comes from m , not from $m - 1$.

7.2 Why the second family works unsigned at two

When m is odd, the normalized parameter $m - 1$ is even. Thus (6.1) immediately proves $v_m(N) \equiv v_m(N/2) \pmod{2^{2R}}$.

When m is even and $R \geq 2$, (6.4) gives the same result, in fact with the extra factor $2^{\nu_2(m)}$. It remains to consider a nonzero even m and $N = 2h$ with h odd. Now $m - 1$ is odd, so the signed theorem says

$$b_{m-1}(2h) + b_{m-1}(h) \in 4\mathbb{Z}.$$

Consequently,

$$\begin{aligned} v_m(2h) - v_m(h) &= m(b_{m-1}(2h) + b_{m-1}(h)) - 2mb_{m-1}(h) \\ &\in 2^{\nu_2(m)+1}\mathbb{Z} \subseteq 4\mathbb{Z}. \end{aligned} \tag{7.1}$$

This proves (1.6) in the last case. The zero parameter has already been separated, so Theorem 1.1 is complete.

7.3 A precise dyadic summary

For the first family, the unsigned parameter-refined bound

$$\nu_2(u_m(N) - u_m(N/2)) \geq 2R + \nu_2(m)$$

holds for nonzero even m , and also for odd m when $R \geq 2$. When m is odd and $R = 1$, replace the difference by a sum.

For the second family, the same parameter-refined bound holds whenever m is odd or $R \geq 2$. At the remaining first-descent case with m even, (7.1) gives the always-valid unsigned exponent $\nu_2(m) + 1$, while the signed statement retains exponent $2 + \nu_2(m)$. The ordinary exponent two holds without exception. The examples in Section 9 show why this distinction cannot simply be suppressed.

8 Framing as a change of coordinate

The preceding proof can be understood without geometric background. Nevertheless, its structure becomes more transparent when the normalized transforms are organized as a family of coordinate changes.

8.1 The additive law for the transformations

For an integral unit A , set

$$\mathfrak{T}_m(A)(y) = A(X_m(y)) =: A_m(y), \quad y = xA(x)^{-m}.$$

If a second transformation has parameter ℓ , its coordinate is

$$z = yA_m(y)^{-\ell} = xA(x)^{-(m+\ell)}.$$

Uniqueness of formal inversion therefore proves

$$\mathfrak{T}_\ell(\mathfrak{T}_m(A)) = \mathfrak{T}_{m+\ell}(A), \quad \mathfrak{T}_0(A) = A. \tag{8.1}$$

Thus $F = \mathfrak{T}_{-1}(A)$ is not an unrelated construction. It is one step in the same integer-parameter family. The shift from b_m to b_{m-1} in (4.8) reflects precisely this additive law.

8.2 The transformed potential

Define

$$\mathcal{W}_m(y) = W(X_m(y)) - \frac{m}{2}H(X_m(y))^2. \quad (8.2)$$

Using $d \log y = (1 - mD(x))d \log x$, differentiation of the right side gives

$$\delta_y \mathcal{W}_m(y) = H(X_m(y)), \quad \delta_y^2 \mathcal{W}_m(y) = \mathcal{B}_m(y).$$

All three series have zero constant term where appropriate, so

$$\mathcal{W}_m(y) = \sum_{n \geq 1} \frac{b_m(n)}{n^2} y^n. \quad (8.3)$$

The signed congruence theorem and Lemma 2.2 now imply an integral dilogarithm expansion

$$\mathcal{W}_m((-1)^m y) = \sum_{d \geq 1} c_{m,d} \operatorname{Li}_2(y^d), \quad (8.4)$$

where

$$c_{m,d} = \frac{1}{d^2} \sum_{e|d} \mu(d/e) (-1)^{me} b_m(e) \in \mathbb{Z}. \quad (8.5)$$

In particular,

$$A_m((-1)^m y) = \prod_{d \geq 1} (1 - y^d)^{-dc_{m,d}}. \quad (8.6)$$

The sign in this coordinate is invisible to odd-prime descent, but it is essential for the all-prime dilogarithm expansion.

This is the framing-integrality interpretation of the coefficient proof. Its general mathematical theory and geometric setting are the subject of [3, 4]. Higher-order congruences under more special framing hypotheses are studied in [5]; no such additional hypothesis is invoked here. In particular, the arithmetic of the input must not be assumed to transfer at every stronger order merely because second order transfers.

9 Examples and limits of stronger assertions

9.1 Low-degree values and exceptional parameters

Direct expansion gives

$$\begin{aligned} A(x) &= 1 + 5x + 49x^2 + 685x^3 + 11807x^4 + 232771x^5 + O(x^6), \\ F(x) &= 1 + 5x + 24x^2 + 200x^3 + 2430x^4 + 36096x^5 + O(x^6). \end{aligned}$$

These values agree with the target entry [1]. The normalized coefficients are already nontrivial polynomials in the parameter:

$$\begin{aligned} b_m(1) &= 5, \\ b_m(2) &= 73 + 50m, \\ b_m(3) &= \frac{2890 + 3285m + 1125m^2}{2}. \end{aligned} \quad (9.1)$$

The third expression has rational polynomial coefficients but takes integer values at every integer m , as guaranteed structurally by (4.3). One should not confuse an integer-valued polynomial with a polynomial all of whose coefficients are integers.

m	$b_m(1)$	$b_m(2)$	$b_m(3)$	$b_m(4)$	$b_m(5)$
-2	5	-27	410	-1915	53130
-1	5	23	365	6343	123255
0	5	73	1445	33001	819005
1	5	123	3650	118059	4015380
2	5	173	6980	301517	13540505
3	5	223	11435	623375	35175630

The cases $m = 0$ and $m = 1$ now have concrete interpretations. The normalized family at $m = 0$ is the Apéry sequence, but the original u_0 and v_0 vanish at positive indices. The reversion identity $v_1(n) = a_n$ is an exact coefficient identity, not just a congruence. Furthermore, $v_2(n) = 2u_1(n)$ for every positive n .

For comparison, some primitive coefficients from (8.5) are

m	$c_{m,1}$	$c_{m,2}$	$c_{m,3}$	$c_{m,4}$
-1	-5	7	-40	395
0	5	17	160	2058
1	-5	32	-405	7371
2	5	42	775	18834

Negative coefficients present no difficulty: integrality, rather than positivity, is the property under consideration.

9.2 The first unsigned family cannot include two unconditionally

At $m = 1$, $N = 2$,

$$u_1(2) - u_1(1) = 123 - 5 = 118 \not\equiv 0 \pmod{4}.$$

Thus a blanket unsigned extension of the first OEIS assertion to $p = 2$ is false. The signed statement instead gives

$$u_1(2) + u_1(1) = 128 \equiv 0 \pmod{4},$$

exactly as predicted by the first-descent proof.

The stronger parameter factor at two also has a genuine boundary for the second family:

$$v_2(2) - v_2(1) = 246 - 10 = 236 \equiv 0 \pmod{4}, \quad 236 \not\equiv 0 \pmod{8}.$$

Since $2R + \nu_2(m) = 3$ in this example, the unsigned parameter-refined bound cannot hold uniformly at two. The original second-order congruence still holds, and the signed sum is 256, divisible by 8.

9.3 Second order cannot uniformly be replaced by third order

A small odd-prime example also limits the possible strengthening:

$$\begin{aligned} u_1(7) - u_1(1) &= 5082313271 \\ &= 7^2 \cdot 103720679, \quad 103720679 \equiv 6 \pmod{7}. \end{aligned} \tag{9.2}$$

The valuation is exactly two, not three. Therefore a universal replacement of p^{2r} by p^{3r} in the first conjecture is false. Because $v_2(n) = 2u_1(n)$ and $7 \nmid 2$, the same example rules out that replacement for the second family as well. This is a counterexample to a uniform higher-order assertion, not a claim that no individual parameters or primes can exhibit stronger divisibility.

10 Exact computation and reproducibility

The companion file `verify.py` uses only the Python standard library and arbitrary-precision integers. Every division that is mathematically supposed to be exact is checked with a remainder test. The general proof above is independent of the computations; the program checks identities, edge cases, and representative congruences by routes that expose indexing or normalization mistakes.

10.1 Coefficient algorithms

For a fixed integer q , write $A(x)^q = \sum_{j \geq 0} h_j x^j$. Logarithmic differentiation gives

$$h_0 = 1, \quad h_j = \frac{q}{j} \sum_{k=1}^j a_k h_{j-k} \quad (j \geq 1). \quad (10.1)$$

The quotient is an integer because A^q is an integral unit. This recurrence is valid for positive, zero, and negative q . To obtain $u_m(N)$, use the *fixed* exponent $q = mN$ throughout the computation through degree N . The exponent must not be changed to mj as the internal coefficient index j varies.

For $m \neq 0$, compute $b_m(N)$ by exact division of h_N by m ; for $m = 0$, use a_N . Then compute $v_m(N) = mb_{m-1}(N)$. This requires $O(N^2)$ integer arithmetic operations and $O(N)$ stored integers for one coefficient at a fixed parameter and target index. These are arithmetic-operation bounds, not unit-cost bit-complexity claims for the large integers involved.

The input numbers are generated directly from the binomial definition using

$$P(n, 0) = 1, \quad P(n, k) = P(n, k-1) \frac{(n-k+1)(n+k)}{k^2}, \quad a_n = \sum_{k=0}^n P(n, k)^2.$$

They are independently compared against the recurrence recorded in A005259 [2]:

$$(n+1)^3 a_{n+1} = (34n^3 + 51n^2 + 27n + 5)a_n - n^3 a_{n-1}, \quad a_0 = 1, \quad a_1 = 5. \quad (10.2)$$

This familiar recurrence is used only as a cross-check; neither its validity nor a proof of it is needed for the congruence argument.

For independent low-degree reversion checks, Lagrange inversion gives

$$[y^n]F(y) = \begin{cases} 5, & n = 1, \\ \frac{1}{1-n} [x^n]A(x)^{1-n}, & n \geq 2. \end{cases}$$

The program verifies $F(y) = A(y/F(y))$, both directions of compositional inversion, and direct polynomial powers of A and F through degree 12. These direct power checks do not assume the formulas $u_m = mb_m$ and $v_m = mb_{m-1}$ as their coefficient-computation method.

10.2 The included run

The archived run used maximum index 250, original parameters $-6 \leq m \leq 6$, and normalized parameters $-7 \leq m \leq 6$. For the transformed congruences it used primes 2, 3, 5, 7, 11, 13 and indices $N = hp^r \leq 250$ with $1 \leq h \leq 6$ and $r \geq 1$. Duplicate prime-index pairs were removed, leaving 71 pairs. This is a specified finite sample, not every possible transformed parameter and index up to 250.

The input Apéry congruences were checked more exhaustively: every prime divisor of every positive $N \leq 250$ was used. The completed run reported **8,142 successful checks**. The grouped counts are:

Check category	Count
Apéry definition versus three-term recurrence	251
Input Apéry second-order congruences	473
Input primitive dilogarithm coefficients	250
Functional equation, two inverse identities, Euler product	4
Direct coefficient identities for u_m and v_m	312
Signed normalized congruences at all selected primes	994
Unsigned normalized congruences in the applicable cases	973
Unsigned v_m congruences, including the prime two	923
Unsigned u_m congruences for odd primes or even m	815
Signed parameter-refined congruences for both families	1704
Odd-prime parameter-refined congruences for both families	1272
Signed transformed primitive dilogarithm coefficients	168
Boundary and higher-order counterexamples	3
Total	8142

The JSON report gives all 71 prime-index pairs, the ungrouped category counts, the exact counterexample values, and the arithmetic environment. The CSV file records sample values of the sequences. No floating-point approximation or probabilistic primality test is used.

10.3 Reproduction commands

From the archive directory, with Python 3.9 or later, run

```
python verify.py --max-index 250 --output-dir verification
```

A larger finite test can be requested, for example, with

```
python verify.py --max-index 500 --parameter-radius 8 \
--multipliers 8 --primes 2 3 5 7 11 13 17 \
--output-dir verification-500
```

Only the default run described above is represented by the archived verification results. To rebuild the article, use

```
latexmk -pdf -interaction=nonstopmode -halt-on-error article.tex
```

The source is also compatible with two ordinary `pdflatex` runs. The computational artifact is not a proof-assistant formalization; the proof is the mathematical argument in this article.

11 Conclusion

The two parameterized assertions in A267220 follow from the same normalized transform, with a shift of one in the framing parameter. The central arithmetic fact is preservation of second-order Gauss congruences after this normalization. Its coefficient proof isolates the only linear term that appears to lose one power of the index and recovers that power by formal integration by parts.

For the stated OEIS conjectures the conclusion is unconditional: every integer parameter, every allowed prime, and every positive pair of indices is covered. In fact, all odd primes work with an additional parameter valuation, and the second family works unsigned at two. The dyadic sign correction is exact, not an artifact of an insufficient estimate, as the smallest counterexample demonstrates. Finally, second order is genuinely the strongest order that can be asserted uniformly for these families without additional restrictions.

A A formal Lagrange inversion proof

For completeness, the coefficient formula used above is established here in the precise formal setting required.

Lemma A.1. *Let $\phi(x) \in 1 + x\mathbb{Q}[[x]]$ and let $X(y)$ be the unique series satisfying $X = y\phi(X)$. For any $G(x) \in \mathbb{Q}[[x]]$ and $n \geq 1$,*

$$[y^n]G(X(y)) = \frac{1}{n}[x^{n-1}]G'(x)\phi(x)^n. \quad (\text{A.1})$$

Proof. Let Res_x select the coefficient of x^{-1} from a formal differential. Residues are unchanged by a substitution $y = x + O(x^2)$. This can be checked on monomials: derivatives have zero residue, and $y'/y = x^{-1}$ plus a power series.

Substitute $y = x/\phi(x)$ in the coefficient residue. Then

$$\begin{aligned} [y^n]G(X(y)) &= \text{Res}_y G(X(y))y^{-n-1} dy \\ &= \text{Res}_x G(x)x^{-n-1}\phi(x)^n \left(1 - \frac{x\phi'(x)}{\phi(x)}\right) dx. \end{aligned}$$

The residue of $d(G(x)x^{-n}\phi(x)^n)$ is zero. Expanding that derivative and rearranging shows that the last display equals

$$\frac{1}{n} \text{Res}_x G'(x)x^{-n}\phi(x)^n dx = \frac{1}{n}[x^{n-1}]G'(x)\phi(x)^n,$$

as required. □

References

- [1] OEIS Foundation Inc., *A267220: Expansion of $\exp(\sum_{n \geq 1} A005259(n)x^n/n)$* . The two supercongruence conjectures are attributed to Peter Bala, October 17, 2024. Entry checked September 20, 2026. <https://oeis.org/A267220>.
- [2] OEIS Foundation Inc., *A005259: Apéry numbers*. Entry checked September 20, 2026. <https://oeis.org/A005259>.
- [3] A. Schwarz, V. Vologodsky, and J. Walcher, *Framing the Di-Logarithm (over $\mathbb{Z}$)*, arXiv:1306.4298 (2013). <https://arxiv.org/abs/1306.4298>.
- [4] A. Schwarz, V. Vologodsky, and J. Walcher, *Integrality of Framing and Geometric Origin of 2-functions (with algebraic coefficients)*, arXiv:1702.07135, version 2 (2017), especially Section 3. <https://arxiv.org/abs/1702.07135>.
- [5] L. Felipe Müller, *Wolstenholme Type Congruences and Framing of Rational 2-Functions*, arXiv:2104.10754 (2021). <https://arxiv.org/abs/2104.10754>.